# Lecture 5. Iterative Methods for Linear Systems

Splitting, spectral radius, Jacobi, Gauss–Seidel, SOR, conjugate gradient

## Contents


**5.1 Direct versus iterative methods** **2**

**5.2 Splitting and convergence** **2**

**5.3 Classical iterations** **3**

**5.4 Conjugate gradient** **4**

**5.5 Selected exercises** **4**

## 5.1 Direct versus iterative methods

**Direct methods** (LU, Cholesky, QR, SVD) have a fixed cost and are accurate; they suit dense matrices of moderate size. **Iterative methods** (Jacobi, Gauss–Seidel, SOR, conjugate gradient, GMRES) produce approximations whose accuracy grows with the iterations; they suit very large, sparse or structured systems, parallel computing and warm starts.

**Example 5.1 (PageRank)**

Let $a_{ij} = 1$ if page $i$ links to page $j$, and $r_i = \sum_j a_{ij}$. A random surfer at a page with $r_i > 0$ follows a random link with probability $p$ and otherwise jumps to a uniform page; from a page without links it jumps uniformly. The transition probabilities are

$$p_{ij} = \frac{1-p}{n} + p\frac{a_{ij}}{r_i} \qquad (a_{ij}/r_i := 1/n \text{ if } r_i = 0),$$

and the scores are the stationary distribution: $x^\top P = x^\top$, i.e. $(I - P^\top)x = 0$. For $n \approx 10^9$, LU takes $\frac{2}{3}(10^9)^3/10^{12}$ seconds, about 20 million years at a teraflop; iterative methods need only products with the sparse link matrix. $I - P^\top$ is never formed: row $i$ of $(I - P^\top)x$ is $x_i - \frac{1-p}{n}\sum_{j:r_j>0} x_j - \frac{1}{n}\sum_{j:r_j=0} x_j - p\sum_{j\to i} x_j/r_j$, the last sum over pages linking to $i$. With the two global sums kept as running totals, a Gauss–Seidel or SOR sweep costs $O(n+\mathrm{nnz}(A))$.

## 5.2 Splitting and convergence

Split $A = M - K$ with $M$ invertible and easy to invert. Then $Ax = b$ iff $x = M^{-1}Kx + M^{-1}b$: a solution is a fixed point of

$$x^{(t+1)} = Rx^{(t)} + c, \qquad R = M^{-1}K, \quad c = M^{-1}b.$$

**Norms.** A vector norm satisfies $\|x\| \geq 0$, $\|x\| = 0 \iff x = 0$, $\|cx\| = |c|\|x\|$ and the triangle inequality (e.g. $\ell_1, \ell_2, \ell_\infty$); a matrix norm is also submultiplicative, $\|AB\| \leq \|A\|\|B\|$. Examples: Frobenius $\|A\|_F = \sqrt{\mathrm{tr}(AA^\top)}$ and the **operator norm** $\|A\| = \sup_{y\neq 0}\|Ay\|/\|y\|$, with

$$\|A\|_1 = \max_j \sum_i |a_{ij}|, \qquad \|A\|_\infty = \max_i \sum_j |a_{ij}|, \qquad \|A\|_2 = \sqrt{\rho(A^\top A)},$$

where $\rho(B) = \max_i|\lambda_i(B)|$ is the **spectral radius**.

If $Ax^\star = b$, then $x^{(t+1)} - x^\star = R(x^{(t)} - x^\star)$, so $\|R\| < 1$ for some operator norm implies convergence.

**Theorem 5.1**

For every operator norm, $\rho(A) \leq \|A\|$. Conversely, for every $\epsilon > 0$ there is an operator norm with $\|A\| \leq \rho(A) + \epsilon$. Hence $\rho(A) = \inf\|A\|$ over operator norms, and $\rho(A) < 1$ iff $\|A\| < 1$ for some operator norm.

**Proof.** If $Av = \lambda v$, $v \neq 0$ (complex in general), then $|\lambda|\|v\| = \|Av\| \leq \|A\|\|v\|$.

Conversely, let $A = STS^{-1}$ (Schur) with $T$ upper triangular, eigenvalues on the diagonal, and $D_\delta = \mathrm{diag}(1, \delta, \ldots, \delta^{n-1})$. Then $T_\delta = D_\delta^{-1}TD_\delta = (SD_\delta)^{-1}A(SD_\delta)$ has entries $\delta^{j-i}t_{ij}$ $(j \geq i)$, with off-diagonal part shrinking as $\delta \to 0$. $\|x\|_\delta = \|(SD_\delta)^{-1}x\|_\infty$ is a vector norm ($M = (SD_\delta)^{-1}$ is invertible, so $\|Mx\|_\infty = 0 \iff x = 0$ and the other axioms transfer). With $y = (SD_\delta)^{-1}x$, its

induced norm is

$$\|A\|_\delta = \sup_{y\neq 0} \frac{\|(SD_\delta)^{-1}A(SD_\delta)y\|_\infty}{\|y\|_\infty} = \|T_\delta\|_\infty = \max_i \sum_{j\geq i} |t_{ij}|\delta^{j-i},$$

which is at most $\rho(A)+\epsilon$ for small $\delta$. ■

**Corollary 5.2**

If $A$ is invertible, the iteration $x^{(t+1)} = Rx^{(t)} + c$ converges to $A^{-1}b$ for every $x^{(0)}$ if and only if $\rho(R) < 1$.

**Proof.** If $\rho(R) < 1$, take $\|R\| < 1$ by Theorem 5.1: $\|x^{(t)} - x^\star\| \leq \|R\|^t \|x^{(0)} - x^\star\| \to 0$ (this direction does not need $A$ invertible). Conversely, if $Rv = \lambda v$ with $|\lambda| \geq 1$, start at $x^{(0)} = x^\star + v$: $\|x^{(t)} - x^\star\| = |\lambda|^t \|v\| \geq \|v\| \neq 0$. ■

## 5.3 Classical iterations

Write $A = L + D + U$ (strictly lower, diagonal, strictly upper parts).

**Jacobi.** $M = D$, $K = -(L+U)$:

$$x_i^{(t+1)} = \frac{1}{a_{ii}}\Big(b_i - \sum_{j\neq i} a_{ij}x_j^{(t)}\Big), \qquad x^{(t+1)} = x^{(t)} + D^{-1}(b - Ax^{(t)}).$$

**Gauss–Seidel.** $M = D + L$ (forward substitution), $K = -U$: new components are used immediately,

$$x_i^{(t+1)} = \frac{1}{a_{ii}}\Big(b_i - \sum_{j<i} a_{ij}x_j^{(t+1)} - \sum_{j>i} a_{ij}x_j^{(t)}\Big).$$

This is coordinate descent for linear systems. In code: writing each new $x_i$ back into $x$ before the next component is Gauss–Seidel; replacing $x$ only after the sweep is Jacobi. Gauss–Seidel needs no second vector but is sequential; Jacobi vectorizes and parallelizes.

**Successive over-relaxation (SOR).** Weight the Gauss–Seidel update against the old value:

$$x_i^{(t+1)} = (1-\omega)x_i^{(t)} + \frac{\omega}{a_{ii}}\Big(b_i - \sum_{j<i} a_{ij}x_j^{(t+1)} - \sum_{j>i} a_{ij}x_j^{(t)}\Big),$$

i.e. $M = \frac{1}{\omega}D + L$, $K = \frac{1-\omega}{\omega}D - U$. $\omega = 1$ is Gauss–Seidel; over-relaxation $\omega > 1$ can accelerate convergence substantially.

**Proposition 5.3**

If $A$ is strictly diagonally dominant by rows, $|a_{ii}| > \sum_{j\neq i}|a_{ij}|$, then the Jacobi and Gauss–Seidel iterations converge for every starting point.

**Proof.** *Jacobi.* $R = -D^{-1}(L+U)$ has $\|R\|_\infty = \max_i \sum_{j\neq i}|a_{ij}|/|a_{ii}| < 1$. *Gauss–Seidel.* Let $Rv = \lambda v$, $R = -(D+L)^{-1}U$, so $-Uv = \lambda(D+L)v$, and take $|v_i| = \|v\|_\infty = 1$. Row $i$ gives $\lambda a_{ii}v_i = -\sum_{j>i} a_{ij}v_j - \lambda\sum_{j<i} a_{ij}v_j$, so

$$|\lambda|\Big(|a_{ii}| - \sum_{j<i}|a_{ij}|\Big) \leq \sum_{j>i}|a_{ij}|,$$

and the right side is less than the bracket by dominance. So $|\lambda| < 1$. ■

Jacobi also converges for strict *column* dominance $|a_{jj}| > \sum_{i\neq j}|a_{ij}|$: $R = -D^{-1}(L+U)$ is similar to $DRD^{-1} = -(L+U)D^{-1}$, whose $\|\cdot\|_1$ is $\max_j \sum_{i\neq j}|a_{ij}|/|a_{jj}| < 1$.

**Example 5.2 (Jacobi for the second-difference matrix)**

Let $A = \text{tridiag}(-1, 2, -1) \in \mathbb{R}^{n\times n}$, with eigenvalues $\lambda_i = 2 - 2\cos\frac{i\pi}{n+1}$ and orthonormal eigenvectors $v_i$. Jacobi has $D = 2I$, $R = I - \frac{1}{2}A$, with the same eigenvectors and eigenvalues

$$\mu_i = 1 - \tfrac{1}{2}\lambda_i = \cos\frac{i\pi}{n+1}, \qquad \rho(R) = \cos\frac{\pi}{n+1} < 1.$$

It converges slowly: $\rho(R) \approx 1 - \frac{\pi^2}{2(n+1)^2}$, so a constant-factor error reduction takes $O(n^2)$ sweeps. With $\varepsilon^{(k)} = \sum_i a_i^{(k)} v_i$, $a_i^{(k)} = \mu_i^k a_i^{(0)}$: each mode decays at rate $|\mu_i|$. Modes near $i = (n+1)/2$ ($\mu_i \approx 0$) die at once; the smoothest ($i = 1$) and most oscillatory ($i = n$, $\mu_n = -\mu_1$) persist. This motivates multigrid, which treats smooth error on coarser grids.

**Cost.** A dense sweep costs $2n^2$ flops, so iteration beats LU only with $o(n)$ sweeps. The gain is for sparse or structured $A$, where $Av$ costs $O(n)$ or $O(n\log n)$ and $A$ need not be stored (e.g. $(Av)_i = v_i - v_{i+1}$).

## 5.4 Conjugate gradient

For symmetric positive definite $A$, $Ax = b$ is equivalent to minimizing $\frac{1}{2}x^\top Ax - b^\top x$ (gradient $Ax - b$). **Conjugate gradient** minimizes along mutually $A$-conjugate directions; in exact arithmetic it terminates in at most $n$ steps, and with preconditioning it converges far sooner. It is the method of choice for huge structured positive definite systems (in a classical plasma physics example: Gauss–Seidel 208,000 iterations, block SOR 765, incomplete-Cholesky-preconditioned CG 25). Details belong to an advanced course.

On the Poisson matrix (sparse, symmetric positive definite, block tridiagonal) Jacobi needs thousands of iterations, Gauss–Seidel and SOR fewer, CG far fewer, all much cheaper than dense Cholesky: *exploit structure*.

## 5.5 Selected exercises

**Motivation.** The eigenvalues of Example 5.2, the SOR range $0 < \omega < 2$, Gauss–Seidel for positive definite systems, and finite termination of conjugate directions.

**Exercise 5.1 (Eigenpairs of the second-difference matrix)**

Let $A = \text{tridiag}(-1, 2, -1) \in \mathbb{R}^{n\times n}$. For $i = 1, \ldots, n$ let $\theta_i = i\pi/(n+1)$ and $v_i \in \mathbb{R}^n$ with entries $(v_i)_j = \sin(j\theta_i)$. Show that $Av_i = (2 - 2\cos\theta_i)v_i$, that these are all the eigenvalues of $A$, that the $v_i$ are mutually orthogonal, and that $A$ is positive definite.

**Solution.** [Supplement] Fix $i$, $\theta = \theta_i$, $s_j = \sin(j\theta)$, $j = 0, \ldots, n+1$. As $s_0 = s_{n+1} = 0$, $(Av_i)_j = -s_{j-1} + 2s_j - s_{j+1}$ for all $j = 1, \ldots, n$, and $\sin((j-1)\theta) + \sin((j+1)\theta) = 2\sin(j\theta)\cos\theta$ gives

$$(Av_i)_j = 2s_j - 2s_j\cos\theta = (2 - 2\cos\theta)s_j.$$

$v_i \neq 0$ as $s_1 = \sin\theta \neq 0$. cos is strictly decreasing on $(0, \pi)$, so the $n$ values $\lambda_i$ are distinct, hence all the eigenvalues, and the $v_i$ are orthogonal ($\lambda_i v_k^\top v_i = v_k^\top A v_i = (Av_k)^\top v_i = \lambda_k v_k^\top v_i$). Finally $\lambda_i > 0$ as $\cos\theta_i < 1$. ■

**Exercise 5.2 (SOR requires $0 < \omega < 2$)**

Let $A$ have nonzero diagonal entries and let $R_\omega = M^{-1}K$ with $M = \frac{1}{\omega}D + L$, $K = \frac{1-\omega}{\omega}D - U$ be the SOR iteration matrix ($\omega \neq 0$). Show that $\rho(R_\omega) \geq |\omega - 1|$, and conclude that SOR converges for every starting point only if $0 < \omega < 2$.

**Solution.** [Supplement] $M$ is lower triangular with diagonal $D/\omega$ and $K$ upper triangular with diagonal $(1-\omega)D/\omega$, so

$$\det R_\omega = \frac{\det K}{\det M} = \frac{\prod_i (1-\omega)a_{ii}/\omega}{\prod_i a_{ii}/\omega} = (1-\omega)^n.$$

Each of the $n$ eigenvalues has modulus at most $\rho(R_\omega)$, so $|1-\omega|^n \leq \rho(R_\omega)^n$. For $\omega \leq 0$ or $\omega \geq 2$, $\rho(R_\omega) \geq 1$, and Corollary 5.2 gives non-convergence for some start. ■

**Exercise 5.3 (Gauss–Seidel converges for positive definite systems)**

Let $A$ be symmetric positive definite. Show that the Gauss–Seidel iteration converges for every starting point.

**Solution.** [Supplement] Here $U = L^\top$, $R = -(D+L)^{-1}L^\top$; by Corollary 5.2 show $|\lambda| < 1$.

*Step 1.* Let $Rv = \lambda v$, $v \in \mathbb{C}^n \setminus \{0\}$, so $-L^\top v = \lambda(D+L)v$. Multiply by $v^*$, with $\delta = v^* D v$, $\sigma = v^* L v$; $L$ real gives $v^* L^\top v = \bar{\sigma}$, so

$$-\bar{\sigma} = \lambda(\delta + \sigma).$$

*Step 2.* $\delta > 0$ as $a_{ii} > 0$. With $v = x + iy$, $v^* A v = x^\top A x + y^\top A y > 0$, i.e. $a := \delta + 2\operatorname{Re}\sigma > 0$. And $\delta + \sigma \neq 0$, else $\bar{\sigma} = 0$, so $\sigma = \delta = 0$.

*Step 3.* With $\sigma = s + it$,

$$|\lambda|^2 = \frac{|\sigma|^2}{|\delta + \sigma|^2} = \frac{s^2 + t^2}{(\delta + s)^2 + t^2},$$

which is $< 1$ iff $s^2 < (\delta + s)^2$ iff $\delta(\delta + 2s) > 0$, true as $\delta > 0$ and $\delta + 2s = a > 0$. ■

**Exercise 5.4 (Conjugate directions terminate in $n$ steps)**

Let $A$ be symmetric positive definite, $\varphi(x) = \frac{1}{2}x^\top A x - b^\top x$, and let $d_0, \ldots, d_{n-1}$ be nonzero and $A$-conjugate: $d_i^\top A d_j = 0$ for $i \neq j$. From any $x_0$ define

$$x_{k+1} = x_k + \alpha_k d_k, \qquad \alpha_k = \frac{d_k^\top r_k}{d_k^\top A d_k}, \qquad r_k = b - A x_k.$$

(a) Show that $d_0, \ldots, d_{n-1}$ are linearly independent.
(b) Show that $\alpha_k$ minimizes $\alpha \mapsto \varphi(x_k + \alpha d_k)$.
(c) Show that $d_j^\top r_k = 0$ for all $j < k$, and conclude that $x_n = A^{-1}b$.

**Solution.** [Supplement]

(a) $\sum_i c_i d_i = 0$ times $d_j^\top A$ gives $c_j d_j^\top A d_j = 0$, so $c_j = 0$.

(b) $\nabla\varphi = -r$, so $\varphi(x_k+\alpha d_k) = \varphi(x_k) - \alpha d_k^\top r_k + \frac{\alpha^2}{2} d_k^\top A d_k$, strictly convex in $\alpha$ with derivative zero at $\alpha = \alpha_k$.

(c) Induction on $k$ ($k = 0$ is vacuous). As $r_{k+1} = r_k - \alpha_k A d_k$,

$$d_k^\top r_{k+1} = d_k^\top r_k - \alpha_k d_k^\top A d_k = 0, \qquad d_j^\top r_{k+1} = d_j^\top r_k - \alpha_k d_j^\top A d_k = d_j^\top r_k = 0 \quad (j < k),$$

by the choice of $\alpha_k$, conjugacy and induction. For $k = n$, $r_n$ is orthogonal to $n$ independent vectors by (a), so $r_n = 0$.

Conjugate gradient generates the $d_k$ on the fly by $A$-conjugating successive residuals, with one product $Ad_k$ per step. ■